AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 30 · STUDENT LESSON

Improve Generalization

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 18: a controlled experiment separates possible causes.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Compare training and validation evidence for model capacity. Try the recall before opening notes.

Your learning target

I can compare **training** and validation **evidence** for model **capacity**.

I can explain **overfitting** without assuming every gap proves it.

I can use a declared selection rule instead of perfect **training** fit.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Overfitting**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Different failure causes require different repairs: simplicity, representative data, regularization, or better features.



Your target

LEARNING TARGET All spotted **training** leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

Builder name and experiment date:

Readiness for the connected explanation

Can a zero **training** error guarantee good validation performance?

Readiness for Distribution shift

Why does a test score need a population description?

My readiness explanation

This week's end goal

By the end, I can explain improve **generalization** using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Improve Generalization case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use improve **generalization** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A more flexible model can represent complicated relationships and accidental details. **Capacity** can help when there is enough relevant **evidence**, but perfect **training** performance is not the goal.

Compare **training** and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, **reference labels** and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **overfitting** and model **capacity**.

Apply the connected idea: Distribution shift

A model trained on one set of conditions may encounter another. Lighting, measuring tools, class frequency and the relationship between **input**s and labels can change.

A shift does not automatically imply failure, but it challenges the assumption that old **evidence** represents current use.

Use fictional classroom data throughout this investigation. Keep the task, units, **reference labels** and data split visible as you reason. Earlier weeks supplied the model-building **process**; this week examines **distribution shift**.

STUDENT LESSON · 3 OF 18

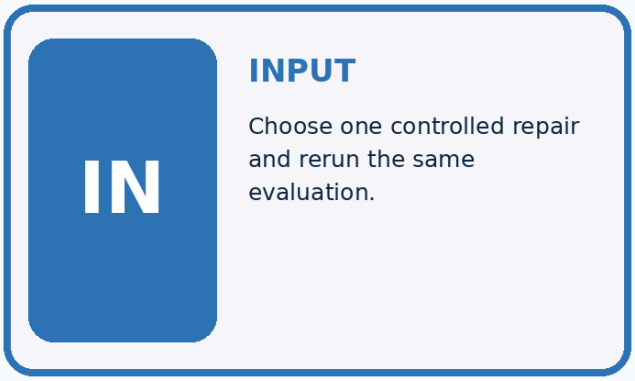
See how it works

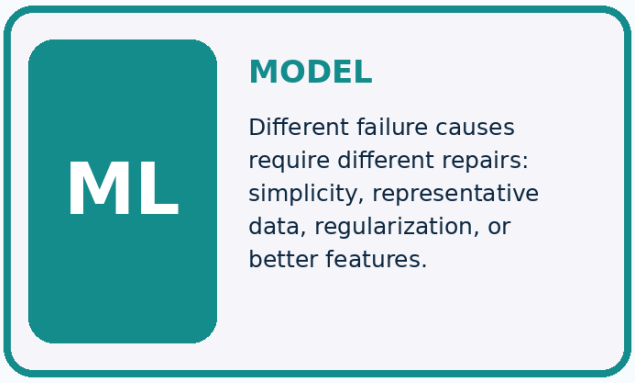
> Follow the mechanism for Improve Generalization. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Choose one controlled repair and rerun the same evaluation.

MODEL IDEA Different failure causes require different repairs: simplicity, representative data, regularization, or better features.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD **Evidence**: high train score, low validation score, largest gap in low light. Hypothesis combines complexity and missing low-light coverage; choose one cause to test

> first.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

On the same regression split, a constant has **training** MAE 5 and validation 6, a line has **training** 2 and validation 3, and a highly flexible curve has **training** 0 and validation 8. Under smallest validation MAE, choose the line.

The curve has the lowest **training** loss but the worst validation result.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Apply the connected idea: Distribution shift

On development-like bright images, a frozen classroom classifier makes 2 errors in 20 cases:10 percent. On a separate dim set it makes 6 errors in 20:30 percent. The pooled result is 8/40=20 percent error.

The pooled score hides a 20-percentage-point difference between conditions.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

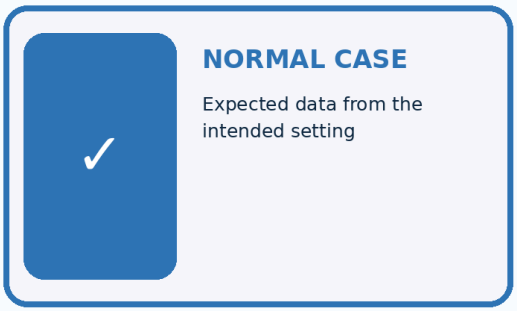
Compare the cases

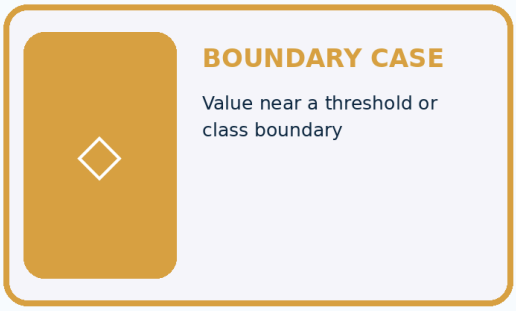
> Compare the supplied cases for Improve Generalization. Identify the detail that changes the conclusion.

EVALUATION DESIGN

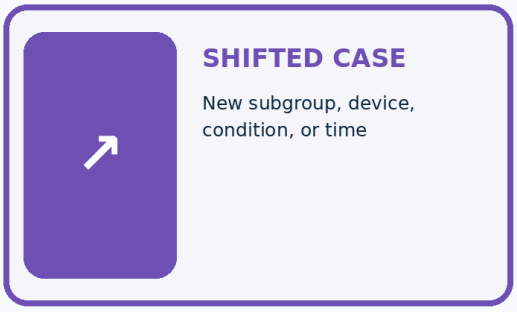
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves **generalization**. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes.

Predicted result

Prediction: It fails on spotted/grey and healthy/red in the crossed audit.

Why the result follows

Reason: Those supplied **prediction**s follow background rather than the intended leaf target.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Improve Generalization to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.


1
GENERALIZATION
performance beyond training examples
2
CONTROLLED CHANGE
one planned modification
3
ABLATION
remove one element to measure its effect
4
REGRESSION TEST
protect previously passing behavior

Term	Precise meaning	My ML example
generalization	performance beyond training examples	
controlled change	one planned modification	
ablation	remove one element to measure its effect	
regression test	protect previously passing behavior	

MISCONCEPTION TO REPAIR Changing data, features, model, and threshold together is the fastest proof of improvement.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Capacity	Flexibility of a model to represent patterns
Overfitting	Learning details that fail to generalize well
Underfitting	Failing to capture useful structure
Generalization	Performance on relevant unseen examples

Terms for Distribution shift

Term	Meaning
Distribution shift	A change in the data-generating conditions
Monitoring	Repeated checks of inputs, outputs and later outcomes
Reference labels	The labels used to evaluate predictions

Term	Meaning
Fallback	A planned alternative when the model is unsuitable

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Improve Generalization, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 30 • WORKED EVIDENCE

1. Evidence: high train score, low validation score, largest gap in low light.
2. Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.
3. Add representative low-light training examples while freezing model settings.
4. Rerun all validation cases; compare overall, error slice, subgroup gap, and regressions.

Step 1

STEP 1 **Evidence**: high train score, low validation score, largest gap in low light.

Step 2

STEP 2 Hypothesis combines complexity and missing low-light coverage; choose one cause to test first.

Step 3

STEP 3 Add representative low-light **training** examples while freezing model settings.

Step 4

STEP 4 Rerun all validation cases; compare overall, error slice, subgroup gap, and regressions.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Overfitting and model capacity at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

Apply the connected idea: Distribution shift

Distribution shift at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Bright condition	2 errors of 20	10 percent error
Dim condition	6 errors of 20	30 percent error
Combined	8 errors of 40	20 percent error

The comparison identifies an observed weakness, not its cause. Check how images and labels were collected and whether other factors changed with lighting. An input shift can be detected before labels arrive, but performance cannot be measured from predictions alone.

Why is 8/40 incomplete by itself? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Improve Generalization

Test whether a plant model follows the leaf or a convenient background shortcut.

The facts we will use

Supplied facts

Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards. Intended use allows both backgrounds for either class.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/30

Improve Generalization

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Notice correlation

Background colour perfectly tracks the labels in this training sample.

2. Design crossed cases

Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved.

3. Inspect fixed model

Compare predictions for the same leaf on both backgrounds; do not assume an error cause before observing it.

CHANGE ONE THING

The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Notice correlation	Background colour perfectly tracks the labels in this training sample.
2. Design crossed cases	Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved.
3. Inspect fixed model	Compare predictions for the same leaf on both backgrounds; do not assume an error cause before observing it.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

Keep the stated limit

Read the labelled picture: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Improve Generalization claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Changing data, features, model, and threshold together is the fastest proof of improvement. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Changing data, features, model, and threshold together is the fastest proof of improvement.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A simpler model is not always better. If a flexible candidate has better valid **evidence** under the declared rule, keep that **evidence** visible.

Model **capacity** is one part of an experiment, alongside features, sample size, label quality and the split.

> **What extra evidence would you inspect before concluding a model memorized noise?**

Apply the connected idea: Distribution shift

Do not normalize away a shift by silently changing pre**process**ing during evaluation. Treat a revised transformation or retrained model as a new version. Use fresh appropriate **evidence** for its claims and retain earlier results.

> **State a safe classroom response to inputs outside the supported range.**

STUDENT LESSON · 8 OF 18

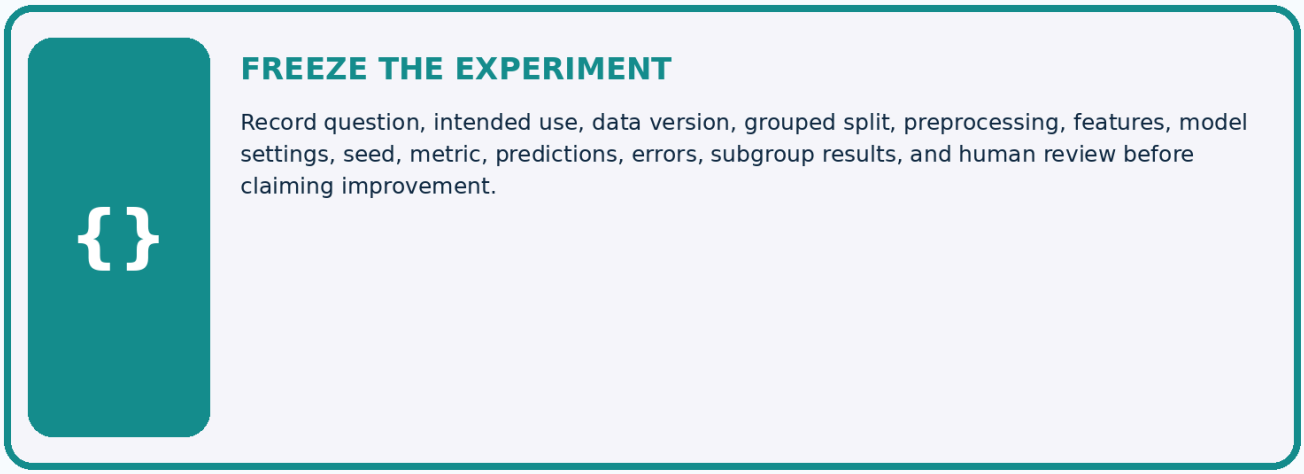
Practise together

> Practise Improve Generalization with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Change one development factor at a time: simplify the candidate, improve data coverage or adjust a declared regularization setting. Refit on **training** and compare on validation.

Do not remove difficult valid examples or repeatedly search the test for a favorable result.

Apply the connected idea: Distribution shift

For a supervised classroom prototype, define a review trigger before use, such as a new unsupported **input** range or a documented increase in labelled error.

Pause the affected use, provide a human or non-model **fallback**, and investigate with new development **evidence**. Version any revision.

Practise with less help: Interpret a gap

Use workbook W2 for the connected example "**Overfitting** and model **capacity**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Distribution shift

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A final answer is enough

Explain the picture

What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

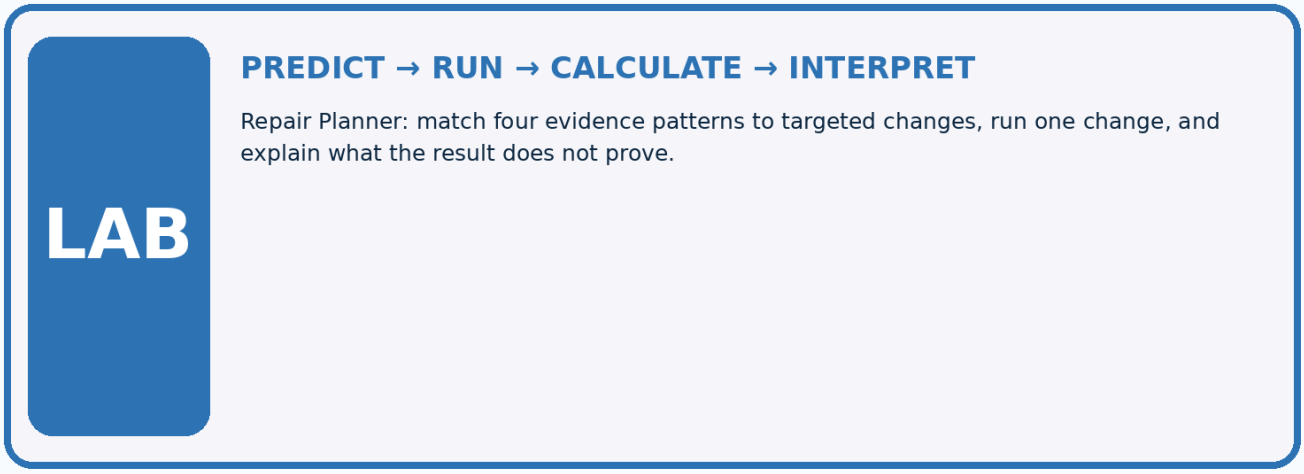
Run the investigation

> Investigate Improve Generalization. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Repair Planner: match four **evidence** patterns to targeted changes, run one change, and explain what the result does not prove.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

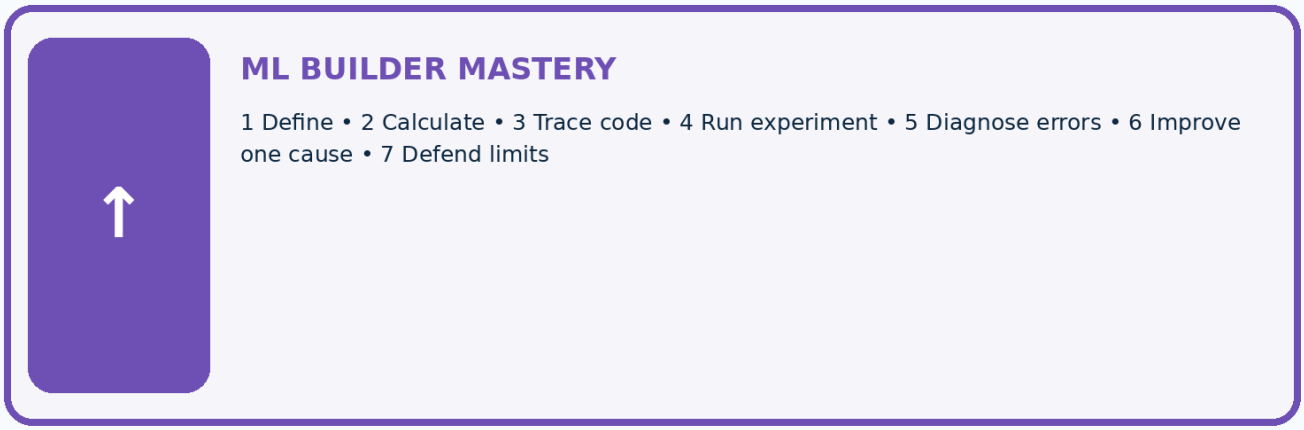
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Improve Generalization. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **generalization**, **controlled change**, **ablation**, **regression test**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Changing data, features, model, and threshold together is the fastest proof of improvement.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Improve Plant Classifier with one approved change and a full before/after **evidence** table.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Choose one controlled repair and rerun the same evaluation.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

> **Connect to the next lesson**
>
> NEXT BRIDGE Carry the experiment **evidence** into: Subgroup Performance.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Improve Generalization case for supported practice. Repeated examples do not become fresh assessments.

Week 30 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A classifier relies on tray colour; new trays reduce performance.

The question

Propose controlled improvement.

Worked explanation

Reduce the shortcut and add representative variation; compare on fixed validation cases including new colours. Change one planned factor.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Repair a background shortcut deliberately

Evidence or stage	Value or result
Observed shortcut	Tray colour tracks label
Changed condition	New trays reduce performance
Revision	Reduce shortcut and add relevant variation
Comparison	Same validation conditions and slices

A classifier relies on tray colour; new trays reduce performance.

Propose controlled improvement.

Reduce the shortcut and add representative variation; compare on fixed validation cases including new colours. Change one planned factor.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Spotted leaves have blue backgrounds and healthy leaves white backgrounds in **training**. Design a generalisation investigation with fresh evaluation.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 30 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A team proposes more representative **training** examples and a simpler model. How should it compare changes while protecting the final test?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Improve Plant Classifier with one approved change and a full before/after **evidence** table.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

All spotted **training** leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Improve Generalization, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Test whether a plant model follows the leaf or a convenient background shortcut.

By the end, I can	How I will show it
Compare training and validation evidence for model capacity	Explain the deciding step, show a trace or test, and state one limit.
Explain overfitting without assuming every gap proves it	Explain the deciding step, show a trace or test, and state one limit.
Use a declared selection rule instead of perfect training fit	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Improve Generalization without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Compare **training** and validation **evidence** for model **capacity**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Test whether a plant model follows the leaf or a convenient background shortcut.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

1. Notice correlation

2. Design crossed cases

3. Inspect fixed model

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect fixed model Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. Read the picture in order.

Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Which conclusion is supported by the supplied facts?

1. It fails on spotted/grey and healthy/red in the crossed audit.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: It fails on spotted/grey and healthy/red in the crossed audit. Those supplied **prediction**s follow background rather than the intended leaf target.

Repair a missing building block

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Overfitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model **capacity**. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B’s zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

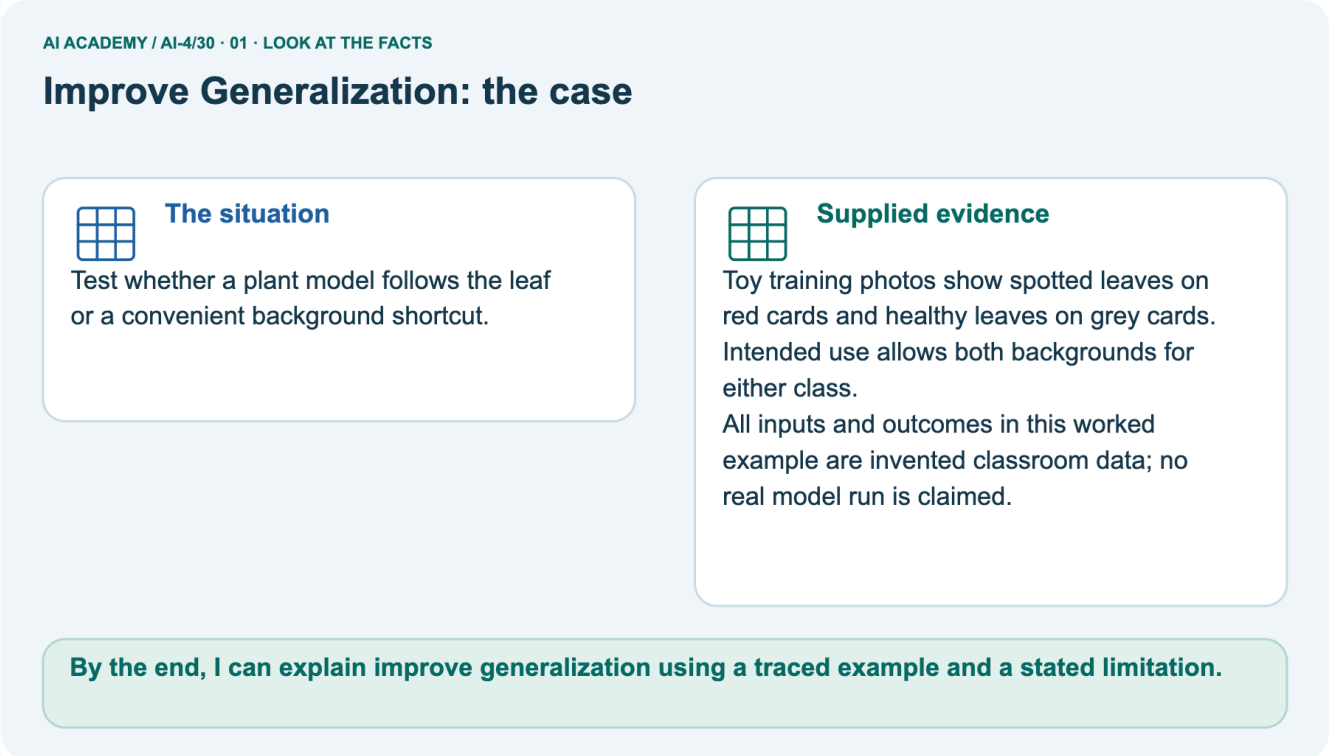
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Improve Generalization graphic using its labels, deciding evidence and relevant limitation.

01. Improve Generalization: the case



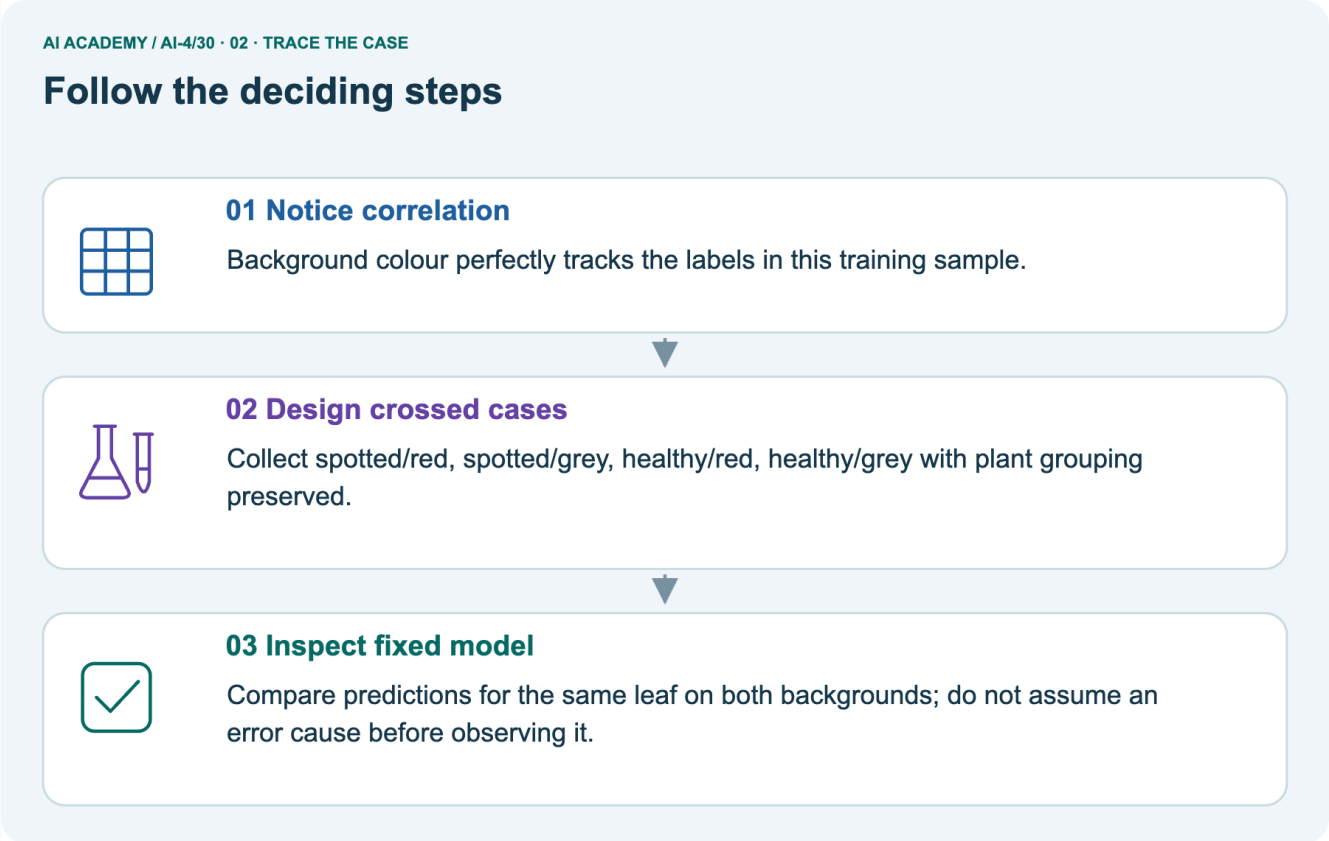
Follow the labelled picture with the matching explanation. Open the vector diagram

Test whether a plant model follows the leaf or a convenient background shortcut. Supplied facts: Toy **training** photos show spotted leaves on red cards and healthy leaves on grey cards.

Intended use allows both backgrounds for either class. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Notice correlation: Background colour perfectly tracks the labels in this **training** sample. → Design crossed cases: Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved. → Inspect fixed model: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Overfitting and model capacity at a glance

AI ACADEMY / AI-4/30 · 03 · INSPECT THE LESSON MODEL

Overfitting and model capacity at a glance

Part	Evidence or action	Interpretation
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

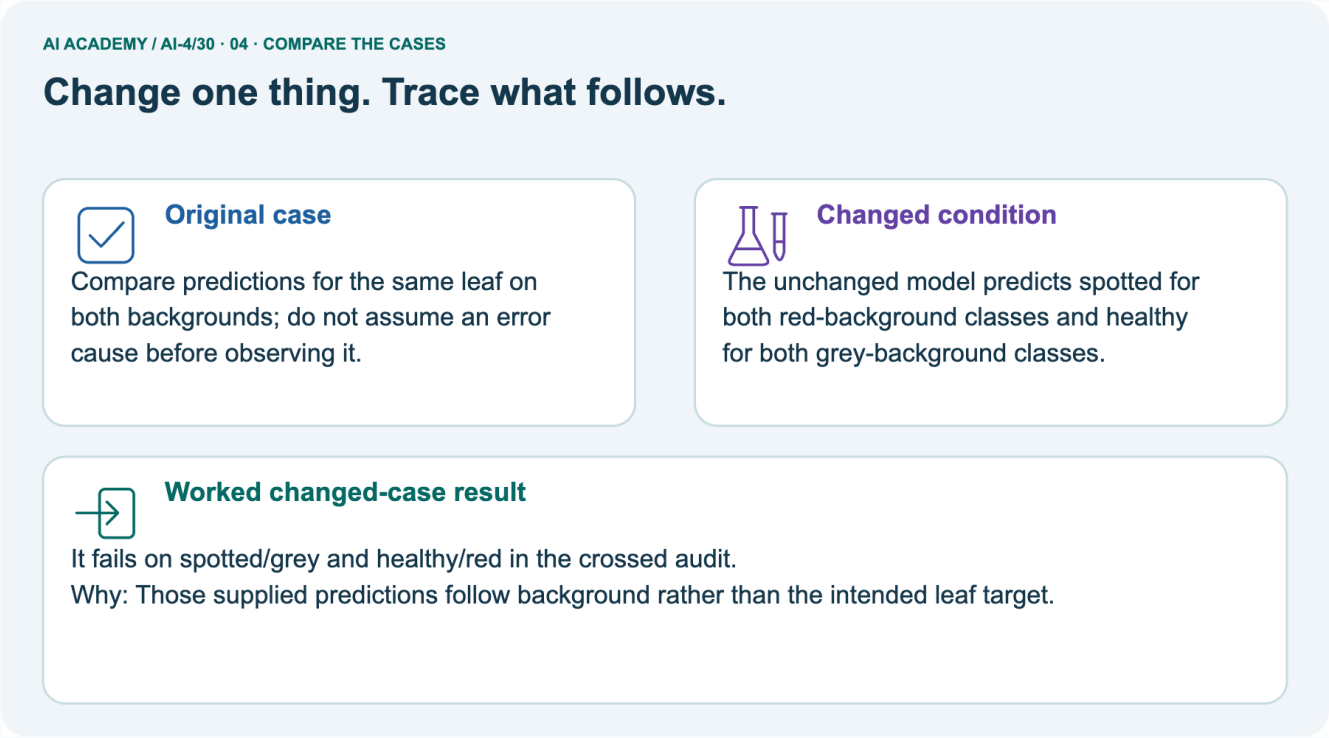
The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Follow the labelled picture with the matching explanation. Open the vector diagram

Overfitting and model **capacity** at a glance. Constant | **Training** 5 | Validation 6. Line | **Training** 2 | Validation 3. Flexible curve | **Training** 0 | Validation 8

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

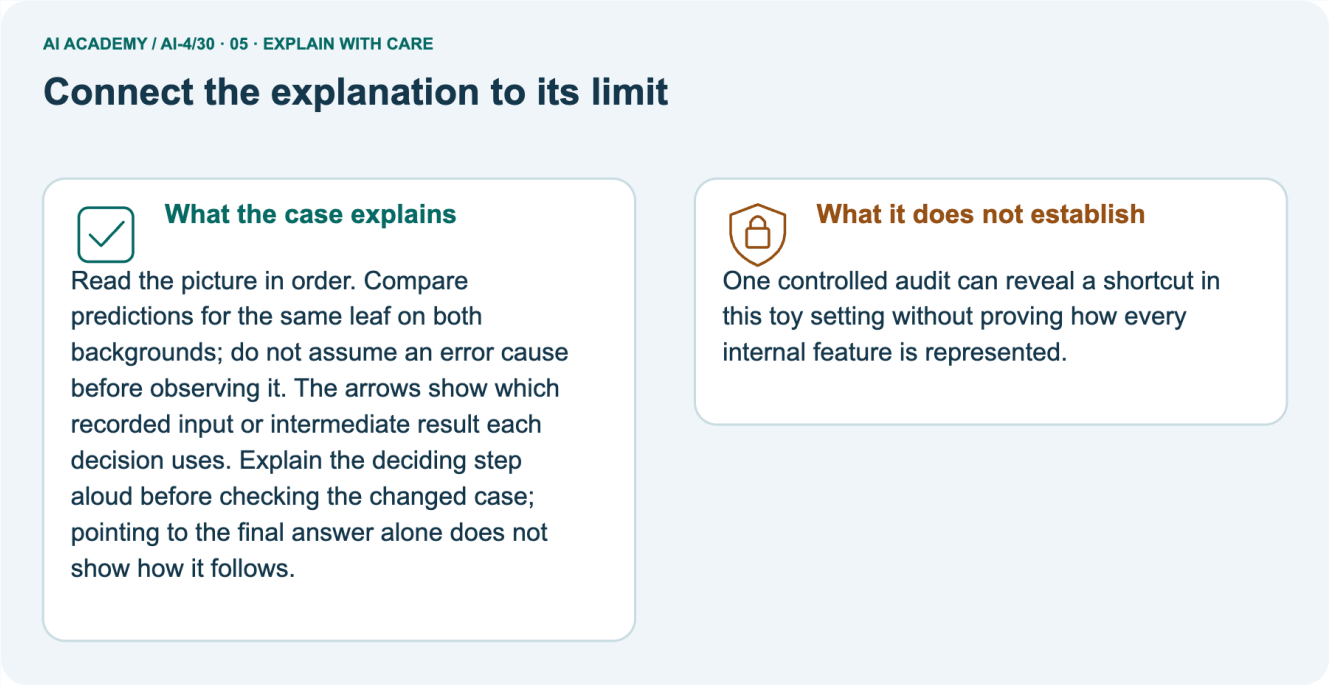
Original case: Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it.

Changed condition: The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes. Expected result: It fails on spotted/grey and healthy/red in the crossed audit.

Reason: Those supplied **prediction**s follow background rather than the intended leaf target.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

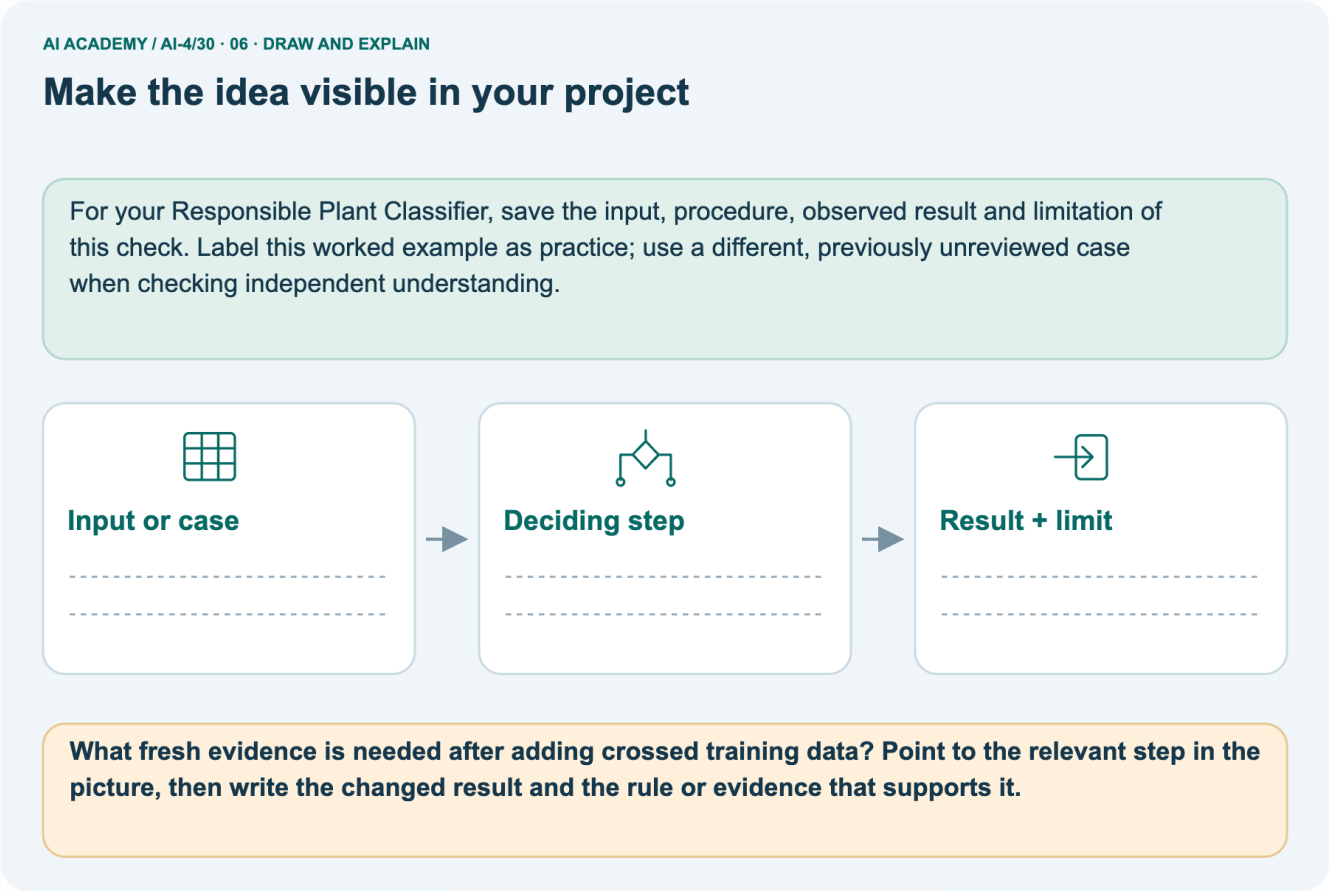
Read the picture in order. Compare **prediction**s for the same leaf on both backgrounds; do not assume an error cause before observing it. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: One controlled audit can reveal a shortcut in this toy setting without proving how every internal feature is represented.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.